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Paul Kelly

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Objective

To continue research and increase my experience in the silicon cybersecurity industry

Education

Rochester Institute of Technology Rochester, NY
Bachelor of Science in Computer Engineering *Graduation: May 2022*

Technical Skills

Languages: C/C++, Rust, Python, ARM and x86 Assembly, VHDL, C#, Java/Smali, Golang
Software: Binary Ninja, Bash, Burp Suite, Windows, Linux, Vim
Technologies: SEM/FIB, FPGA, ESP32, SDR, Bluetooth and BLE, 802.1x, 3G/4G/5G Cellular, Signal Analyzer

Experience

IOActive, Inc Seattle, WA
Senior Security Consultant *June 2022 - Current*

- Silicon - Perform decapping, delayering, and netlist extractions on integrated circuits to identify vulnerabilities or stolen IP
- Automotive - Total vehicle assessments to identify weaknesses in inter-module communications, gateway routing, or general packet handling
- Medical - Security assessments spanning the entire industry of new-age medical equipment including MRI machines, Ultrasound scanners, X-rays, and surgical robots
- Cellular - Utilize SDR to run 3G/4G/5G base stations in the field to assess security posture from rogue base stations.
- Reverse Engineering - Aided by experience with broad range of binary file formats and ISAs, both standard and custom
- Threat Modeling - Performed threat modeling of embedded/automotive systems using STRIDE and attack-surface analysis, identifying trust-boundary violations, exploitable attack paths, and security controls across firmware, diagnostic interfaces, and network protocols.
- Exploit Development - Python and C programs to take advantage of stack overflows, format string bugs, object deserialization, and more

Saab Defense and Security Syracuse, NY
Electrical Engineer Intern *May 2019 - Dec 2019*

- Ran validation processes for RF hardware including writing VHDL test benches, algorithm confirmation MATLAB scripts, and instrumented test equipment to generate mixer intermodulation tables and analyze harmonics.

Projects

Silicon ROM Extraction
- Decapped, delayered (through polishing and ion beam milling), and imaged region of interest to rebuild bit mask ROM into byte code that can be reverse engineered through traditional means

Silicon Counterfeits
- Decapped, delayered, and imaged silicon chips to extract netlists and identify stolen logic blocks from copyrighted designs

MIPS Processor Model
- Designed the instruction fetch, instruction decode, execute, memory access, and write back stages of the MIPS processor in VHDL

Links
- Github: https://github.com/pkelly916/
- Website: http://pauljkelly.net